It is reasonable to assume, as de Broglie did in 1924, that since photons can behave as moving particles moving particles may show wave properties. From the previous equations, we can — by replacing **c** by $v$ — derive that the wavelength of such *matter waves* is

$$\lambda = \mathbf{h}/mv \tag{2.7}$$

This relation is of importance in explaining nuclear reactions, and has led to practical consequences in the use of electron diffraction, and in the development of electron microscopy.

The wave and particle properties of matter complement each other (the *complementarity principle*; N. Bohr, 1928). Throughout this book we either use models based on wave properties or sometimes on particle properties, depending on which more directly explain the particular phenomenon under discussion.

## 2.4. FORMATION AND PROPERTIES OF SOME ELEMENTARY PARTICLES

The track formed by a moving particle in the magnetic field of a bubble chamber is characterized by its width, length and curvature. From a kinematic analyses of the tracks it is possible to determine the mass and charge of the particles involved, see Ch. 3, and Ch. 9. Further, as seen at points A-B and E-F in Figure 2.2, the interruption of a track can indicate formation of an uncharged particle, as they do not form visible tracks. From a knowledge of the tracks formed by known particles, "strange tracks" can be analyzed to identify new particles in bubble chamber pictures and to assign their properties. All these properties have to be quantized, so new quantum states have been introduced, like baryon number, statistics, symmetry, parity, hypercharge, isospin, strangeness, color, etc. in addition to spin. It is found that many of the particles observed are sensitive to only one or two of the forces of nature (§2.2), which serves as an additional aid in their classification. Nevertheless, the array of properties assigned to the hundreds of "elementary particles" which have been discovered resembles the situation when 40 radioelements were reported between uranium and lead, as described in chapter 1.

Before we proceed to the order which has evolved from this picture we must describe some of the concepts used to define these particles. It has been practical to divide the particles according to their masses: (i) *baryons* are the "heavy" ones (protons, neutrons, hyperons and nuclei), (ii) *leptons* are the "light" ones (the electron, the neutrino and the muon); (iii) *mesons* have "intermediate" masses; these include the π-meson, the K-meson, etc. The baryons and the mesons have also been considered as *hadrons*, "hard" or "strong" particles as they take part in the strong nuclear force. Such properties were used to develop a table of elementary particles, Table 2.2.

**Table 2.2** Common "elementary particles" in nuclide reactions

| Class | Name | Symbol | Rest mass (MeV) | Rest mass (u) | Life-time (s); decay mode |
|---|---|---|---|---|---|
| **Baryons**[†] (nucleons): S, A-yes, P+, s = ½ | neutron | n | 939.6 | 1.008665 | 889; $n \rightarrow p + e^- + \bar{\nu}_e$ |
| | proton | p | 938.3 | 1.007276 | stable |
| **Mesons**: S, A-no, P−, s = 0 | K-meson (kaon) | $K^+$, $K^-$ | 493.7 | 0.530009 | $\approx 10^{-8}$; $K^+ \rightarrow \mu^+ + \nu_\mu$, $\pi^+ + \pi^0$, etc.[††] |
| | π-meson (pion) | $\pi^+$, $\pi^0$, $\pi^-$ | 139.6[†††] | 0.149867 | $\approx 10^{-8}$; $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$[††] |
| **Leptons**[††††]: W, A-no, P-no, s = ½ | muon | $\mu^-$ | 105.6 | 0.113366 | $2\times10^6$; $\mu^- \rightarrow e^- + \nu_e + \nu_\mu$ |
| | electron stable | $e^-$ | 0.5110 | 0.0005486 | |
| | neutrino | $\nu_e$, $\nu_\mu$, $\nu_\tau$ | 0? | 0? | stable |
| **Photon**: W, A-no, P-no, s = 1 | photon, gamma | hν<br>γ | 0 | 0 | stable |

S = Strong interaction; W = Weak interaction; A-yes = baryon number conserved; P± = conservation of parity and parity sign; P-no = parity not conserved; s = spin quantum number. All these particles have their anti-particles, except for the photon and mesons, which are their own anti-particles.

[†]Baryons include the nucleons (n and p), hyperons (also called "strange baryons", with $\mathbf{m}_0$ 1116 for $\Lambda^0$ to 1673 MeV for $\Omega^-$, and lifetimes $\approx 10^{-10}$ s), and all nuclei.

[††]$K^-$ and $\pi^+$ decay into corresponding anti-particles in a similar way.

[†††]$\pi^0$ rest mass is 135.0 MeV.

[††††]See also Table 10.2.

All the particles in Table 2.2 have *spin*[2]. Quantum mechanical calculations and experimental observations have shown that each particle has a fixed spin energy which is determined by the *spin quantum number s* ($s = \pm ½$ for leptons and nucleons). Particles of non-integral spin are called *fermions* because they obey the statistical rules devised by Fermi and Dirac, which state that two such particles cannot exist in the same closed system (nucleus or electron shell) having all quantum numbers the same (referred to as *the Pauli principle*). Fermions can be created and destroyed only in conjunction with an anti-particle of the same class. For example if an electron is emitted in β-decay it must be accompanied by the creation of an anti-neutrino. Conversely, if a positron $\beta^+$ − which is an anti-electron − is emitted in the β-decay, it is accompanied by the creation of a neutrino.

Fermions are the *building blocks* of nature. There is another group of "particles", called *bosons*, to which the photon and mesons belong. The bosons are the *carriers of forces*. When two fermions interact they continually emit and absorb bosons. The bosons have an even spin (0, 1, etc), they do not obey the Pauli principle, and they do not require the formation of anti-particles in their reactions.

The well known Schrödinger wave equation is given by eqn (2.8). It requires that the mass of e.g. an electron is constant and independent of its velocity.

$$\mathrm{i}\,\mathbf{\hbar}\frac{\partial\psi}{\partial t} = \widehat{\mathrm{H}}\psi \tag{2.8}$$

However, the well tested Lorenz equation (2.9) shows that this is not true in our universe.

$$m_v = \boldsymbol{m}_0/\left(1 - (v/\mathbf{c})^2\right)^{½} \tag{2.9}$$

Here $m_v$ is the electron mass at velocity $v$, $\boldsymbol{m}_0$ is the rest mass, and $\mathbf{c}$ is the velocity of light. Hence there was a search for an alternative wave equation which would be valid in our universe with its limited speed of light. In 1928 Paul Dirac proposed his well known alternative (2.10) to the Schrödinger equation.

$$\left(\beta\boldsymbol{m}_0\mathbf{c}^2 + \sum_{k=1}^{3}\alpha_k p_k\mathbf{c}\right) \times \psi(x,t) = \mathrm{i}\,\mathbf{\hbar}\frac{\partial\psi(x,t)}{\partial t} \tag{2.10}$$

where $\boldsymbol{m}_0$ is the rest mass of the electron, $\mathbf{c}$ is the speed of light, $p$ is the momentum, understood to be an operator in the sense of the Schrödinger theory, $x$ and $t$ are the space and time coordinates and $\mathbf{\hbar}$ is the reduced Planck constant, $\mathbf{h}$ divided by $2\pi$, β and $\alpha_k$ are

[2] Spin is an intrinsic property of elementary particles, sometimes wrongly thought of as a rotation.

$4 \times 4$ matrices, and $\psi$ is the wave function which has four components. This equation is valid also when the speed of light is finite and it is thus relativistically correct.

The analytical solution of the Dirac wave equation for the energy a single electron is enclosed in a square root. Normally the negative sign is neglected for many such solutions. However, Dirac later recognized that also the solution with negative energy could have a physical meaning. The *hole theory* assumes that the negative energy levels of our universe are completely filled with all kind of particles at all possible quantum states. If sufficient energy is added (> twice the rest mass of the particle) to a point in space it is possible to "lift" such a particle from the negative part of our universe into our normal space. Behind it is left a hole, which to us looks like the same particle but with opposite charge − an anti-particle. A similar theory is used to explain the electrical behavior of semiconductors, see Ch 9.

When the hole theory is generalized, it follows that all particles have their corresponding anti-particles, hence all the particles mentioned have their *antiparticles* (designated by a bar above the particle symbol), except the photon and the mesons, who are their own antiparticles. We may think about *antimatter* as consisting of antiprotons and antineutrons in an antinucleus surrounded by antielectrons (i.e. positrons). Superficially, there would be no way to distinguish such antimatter from our matter (sometimes called *koino matter*). It has been proposed that the universe is made up of matter and antimatter as a requirement of the principle of *symmetry*. In that case some galaxies, which perhaps can be observed, should be made up of antimatter. When such antimatter galaxies (or material expelled from them) collide with koino matter galaxies, both types of matter are annihilated and tremendous amounts of energy released.

Some of the problems with the Dirac hole theory, like a possibly unlimited amount of particles with negative energy, is addressed by the Quantum Field Theory (QFT). However, that theory also seems to have its problems like an infinite amount of vacuum energy. QFT will not be used here and the interested reader is referred to modern text books in particle physics.

In order to reach the goal of a comprehensive, yet simple, theory of the composition of all matter, the properties of the neutrinos and the quark theory must be considered.

## 2.5. THE NEUTRINO

The neutrino is a fermion that plays an essential role in the models of elementary particles and in the theory of the formation and development of the universe. The existence of the electron neutrino was predicted by Pauli in 1927 but it was not proven until 1956 when Reines and Cowan detected them in experiments at the Savannah River (USA) nuclear reactor. Since neutrinos are emitted in the $\beta^-$ decays following fission, nuclear reactors are the most intense neutrino sources on earth. The detector in the discovery experiments consisted of a scintillating solution containing cadmium surrounded by